

Advanced Animal Movement Modelling Training Workshop

Environmental Data Access and Extraction

17 September 2026

Josh Cullen



Stellenbosch
UNIVERSITY
IYUNIVESITHI
UNIVERSITEIT



NITheCS
National Institute for
Theoretical and Computational Sciences

UC SANTA CRUZ

Photo credit: Larry Li

What is remote sensing?

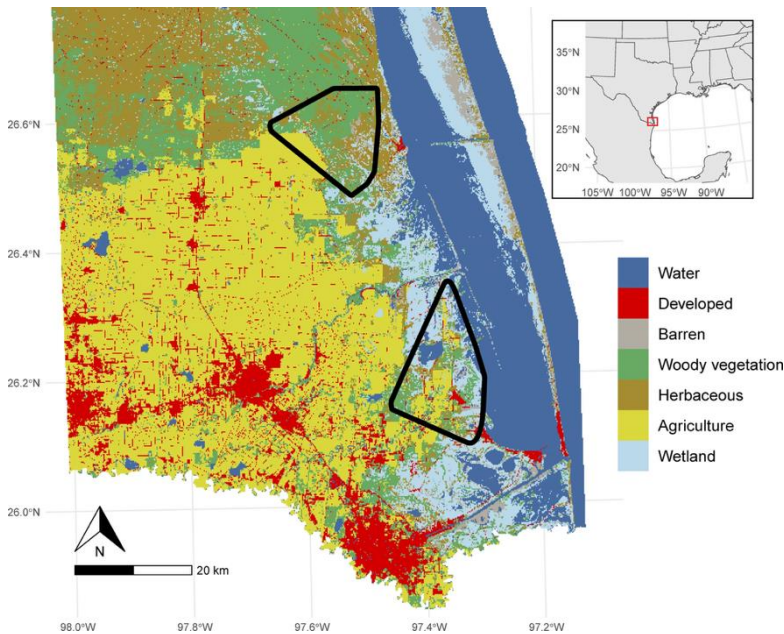
Per NASA:

- “The acquiring of information from a distance...via remote instruments on space-based platforms (e.g., satellites or spacecraft) and on aircraft that detect and record reflected or emitted energy.”

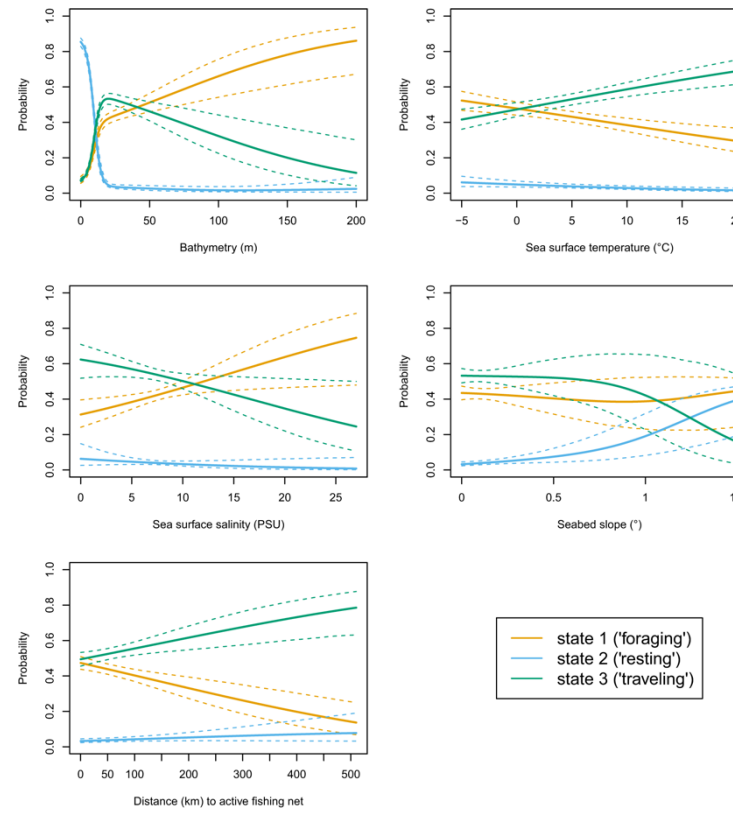


<https://www.earthdata.nasa.gov/data/projects/arset>

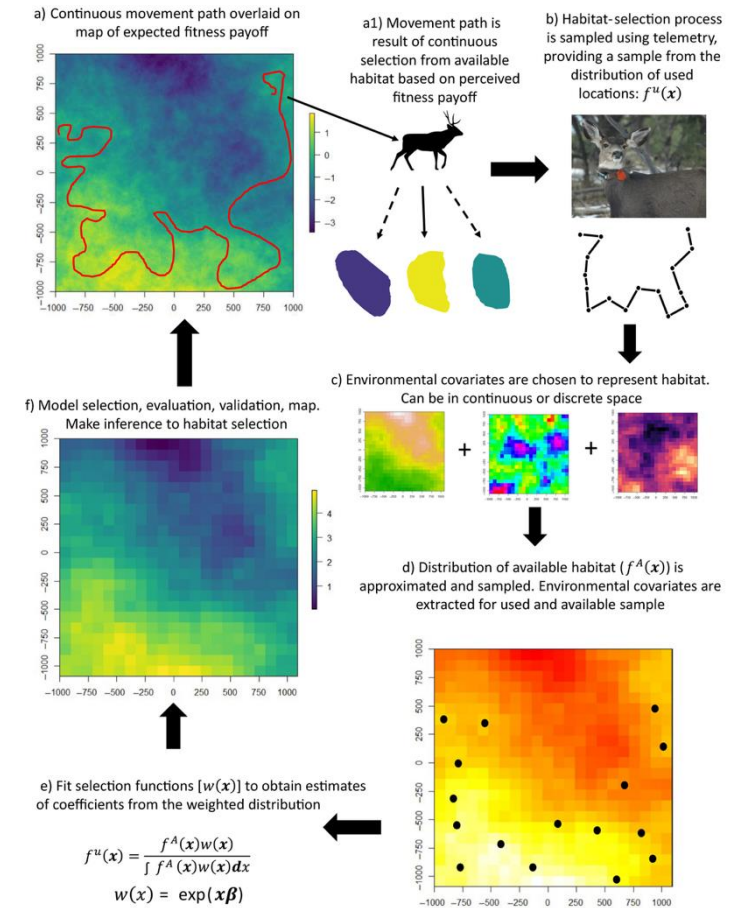
Uses for environmental data in movement ecology



Smith et al. 2025



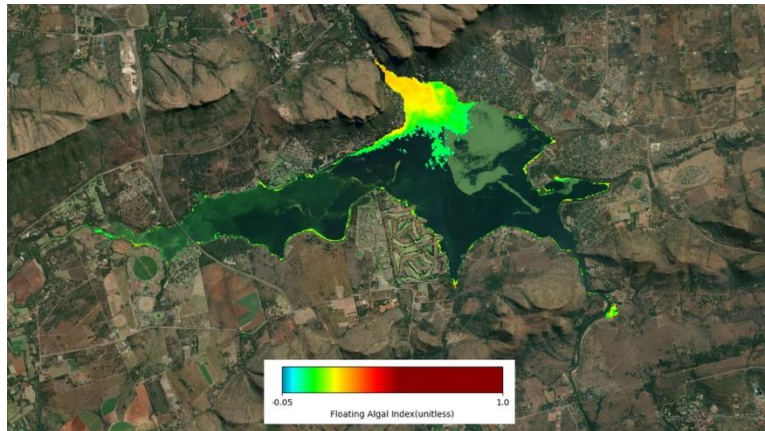
van Beest et al. 2019



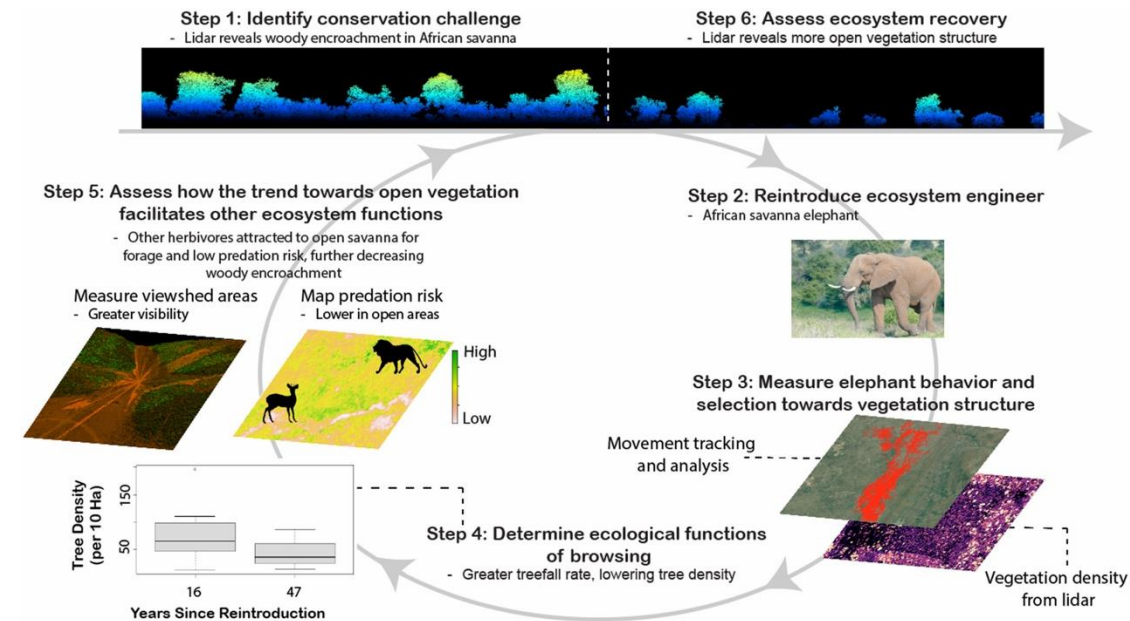
Northrup et al. 2022

Remote sensing for ecosystem restoration

- By monitoring the landscape and animal movement use simultaneously, we can better understand the feedback between them
- It is useful to understand how animals move and select habitat in undisturbed ranges to better inform restoration of habitat and landscape connectivity



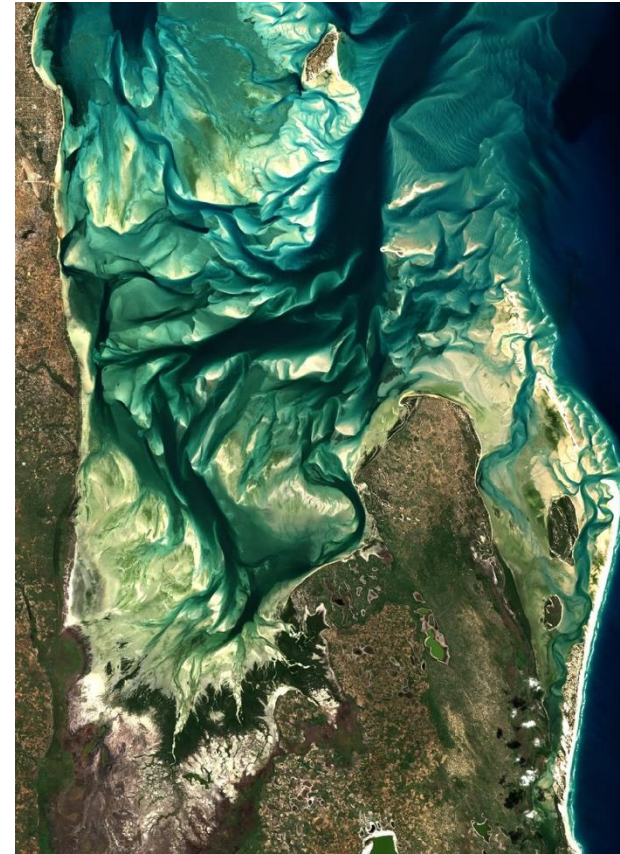
Digital Earth Africa



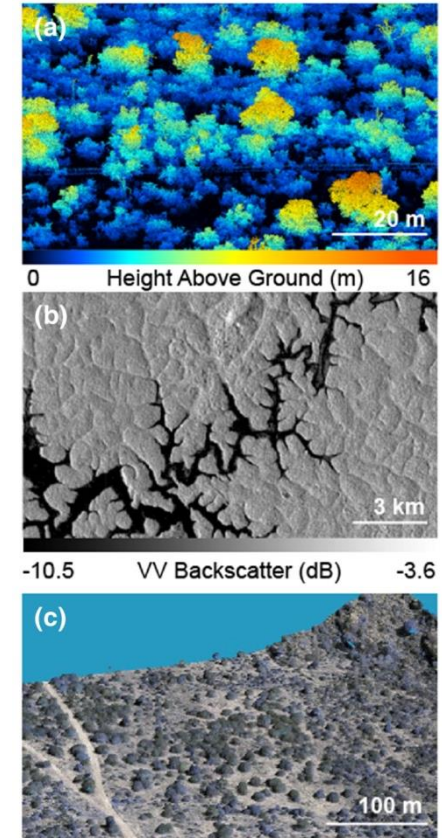
Russo et al. 2023

Types of remote sensing data

- **Optical**
- SAR (synthetic aperture radar)
- LiDAR (light detection and ranging)
- **Thermal infrared**
- Passive microwave
- Nighttime lights
- **Radar Altimetry**
- Hyperspectral



Digital Earth Africa



Russo et al. 2023

Optical

- Essentially images that collect data on bands at different wavelengths (visible and invisible)



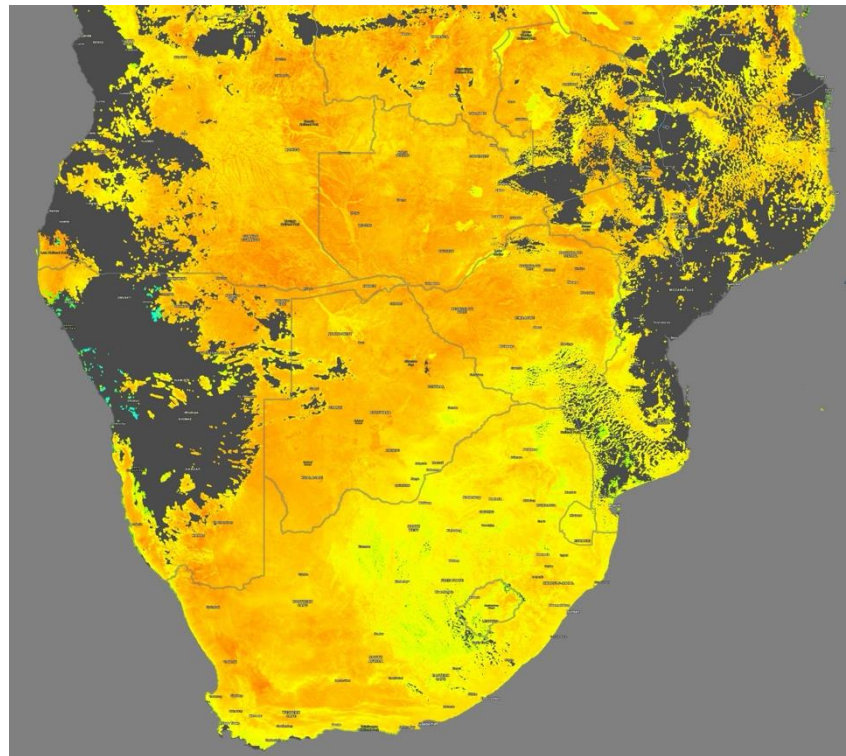
Esri Sentinel-2 Explorer



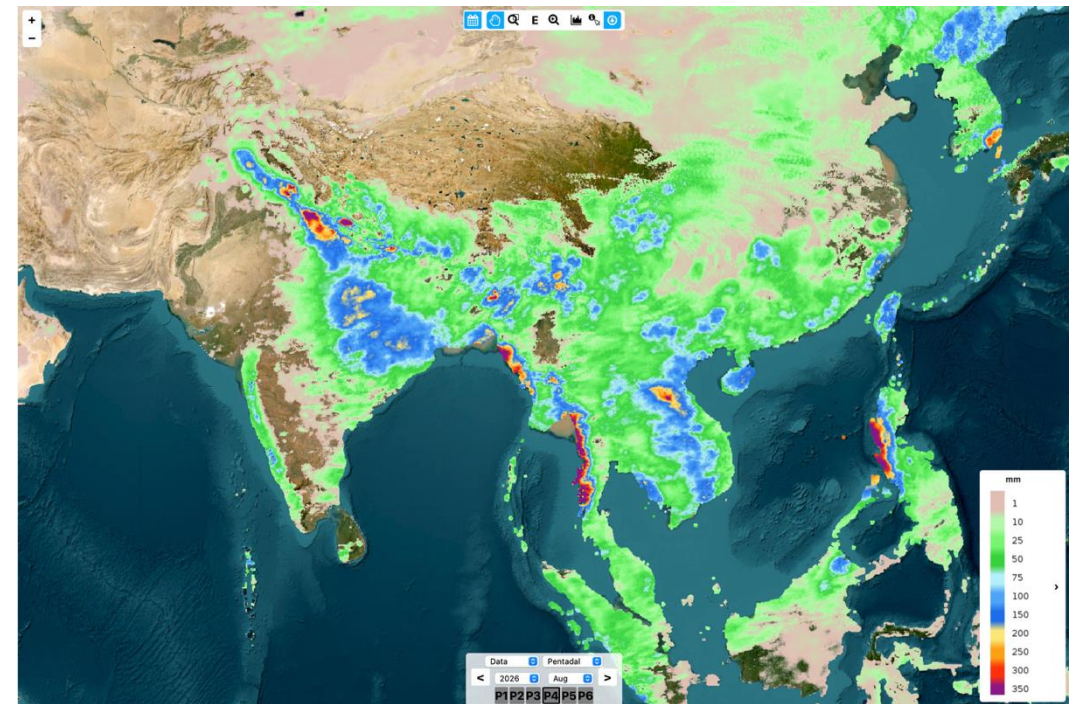
Esri Landsat Explorer

Thermal infrared

- Can be used to measure variables like land surface temperature and rainfall



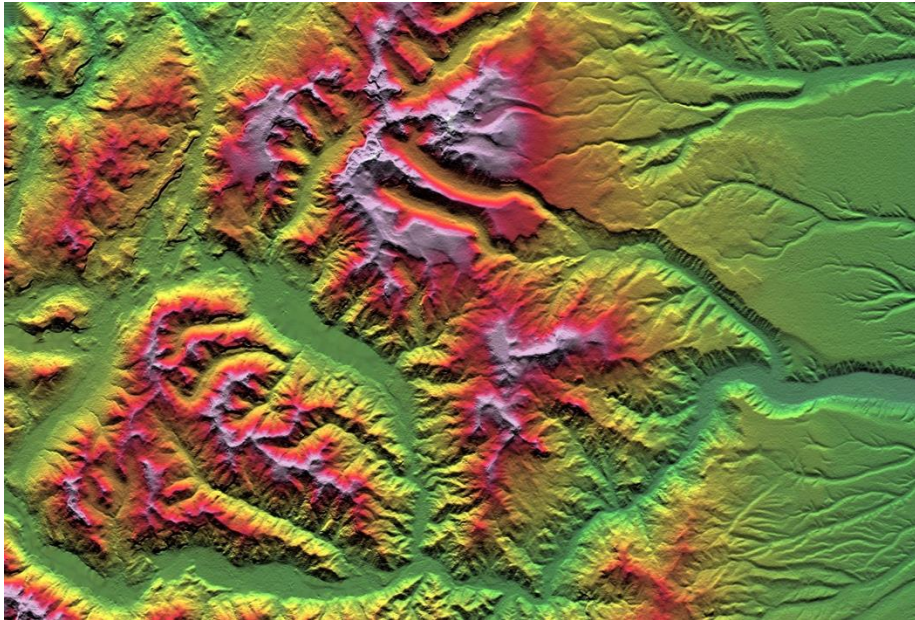
NASA/NOAA Suomi NPP, NOAA-20; NASA Worldview



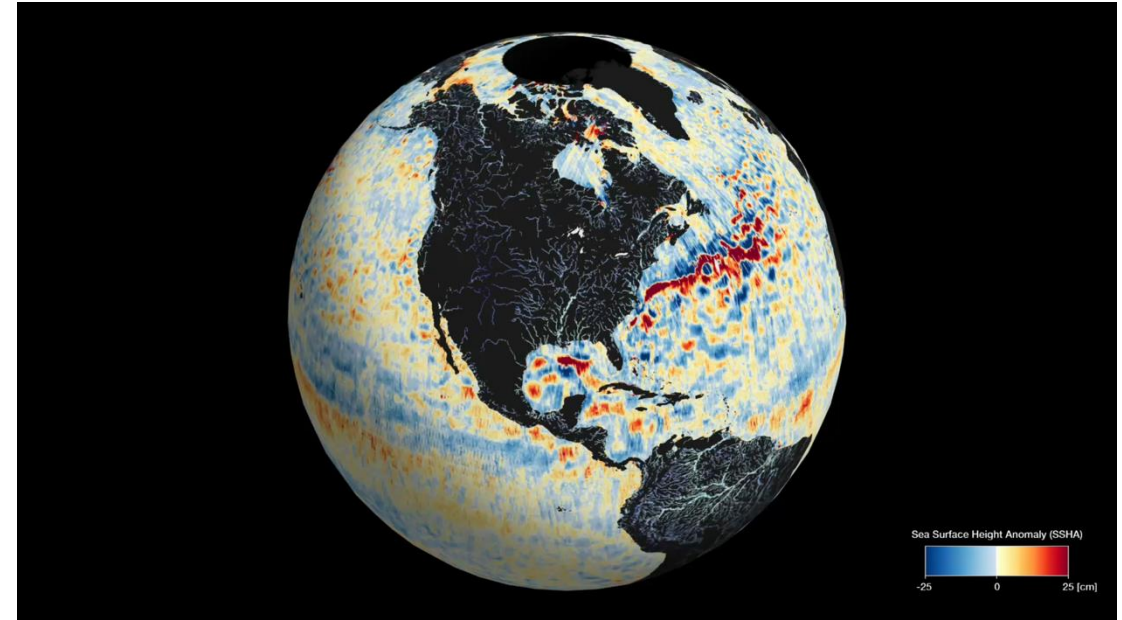
CHIRPS v3; USGS EWX

Radar altimetry

- Measures height of land and ocean surface



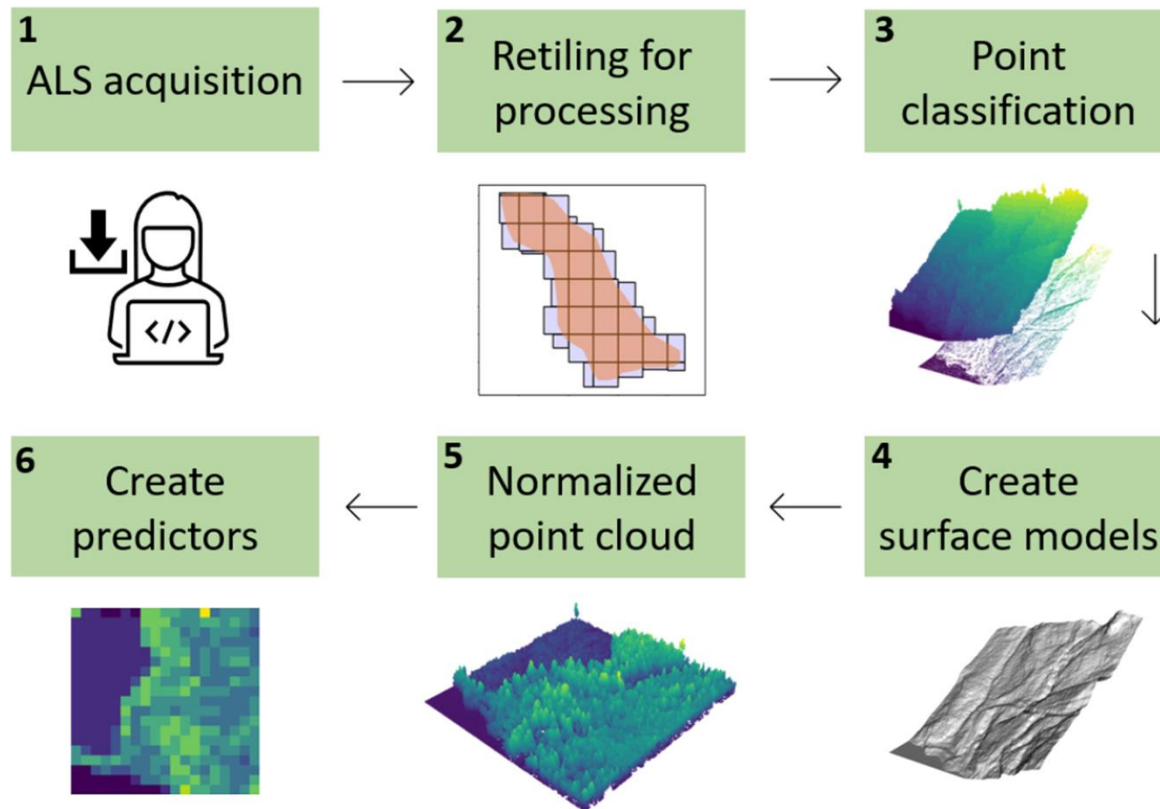
NASA, JPL, NIMA; SRTM



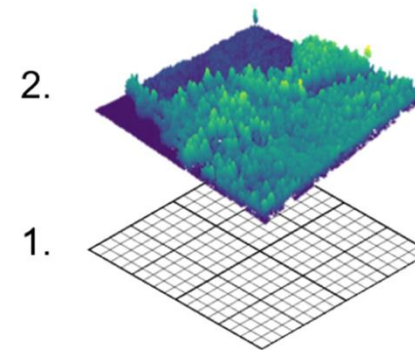
NASA, SWOT, Scientific Visualization Studio

LiDAR

- High-resolution and 3D data
 - Includes airborne laser scanning (ALS)

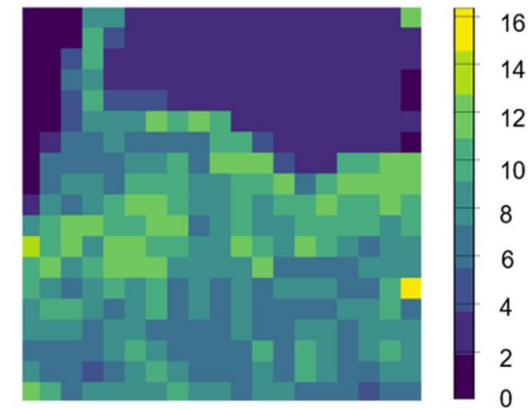


Metric visualization



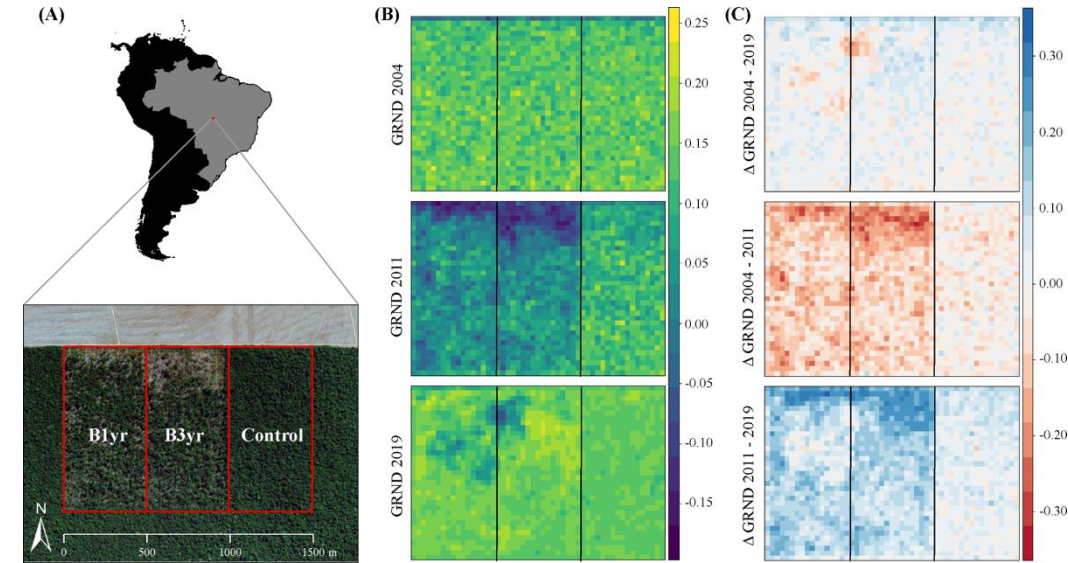
Example output

Vegetation height (m)

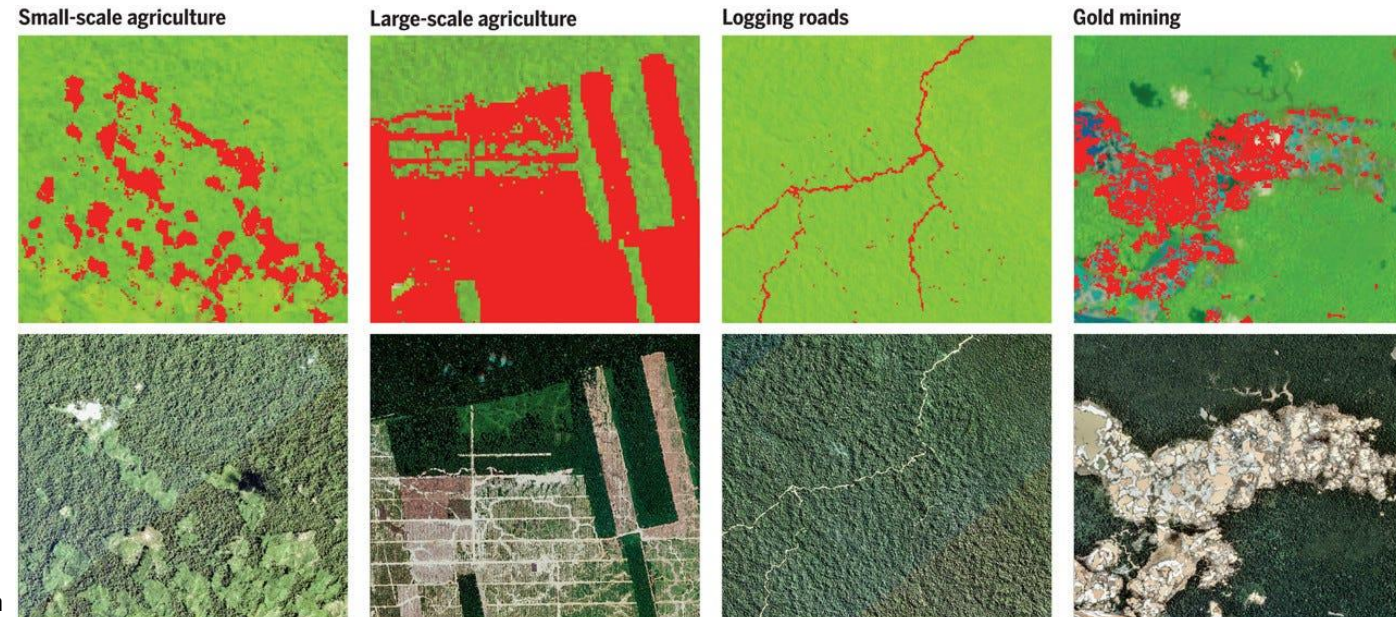


Disturbance detection

- Using one or more of these different satellite products, we can evaluate when a change has happened
 - Deforestation
 - Wildfire
 - Land cover change



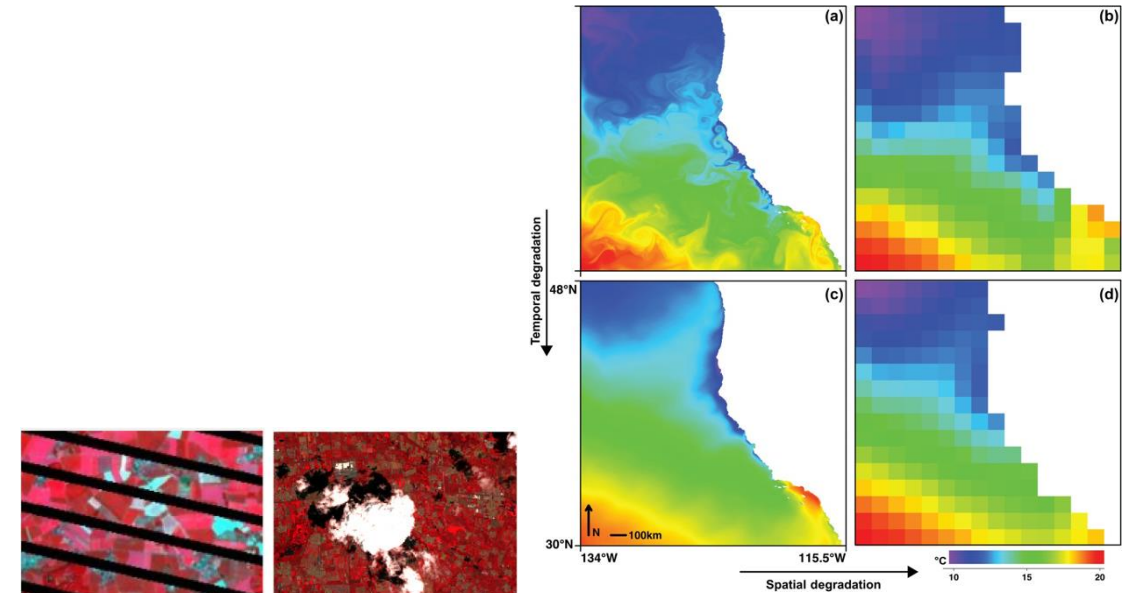
Marques et al. 2024



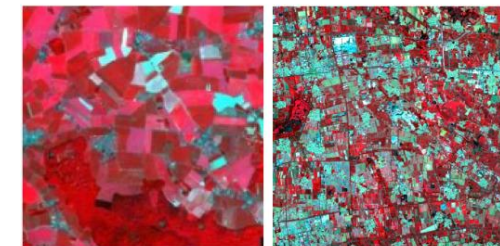
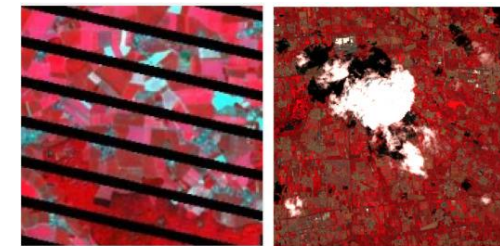
Credit: Hein de Wilde, Medium

Considerations when selecting datasets

- **Spatial and temporal grain** of data related to animal movement and study objectives
- **Static versus dynamic covariates** that explain when, where, and how animals interact with their environment
- **Missing data and cloud cover** may limit when and where data may be available



Scales et al. 2017



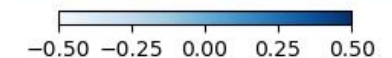
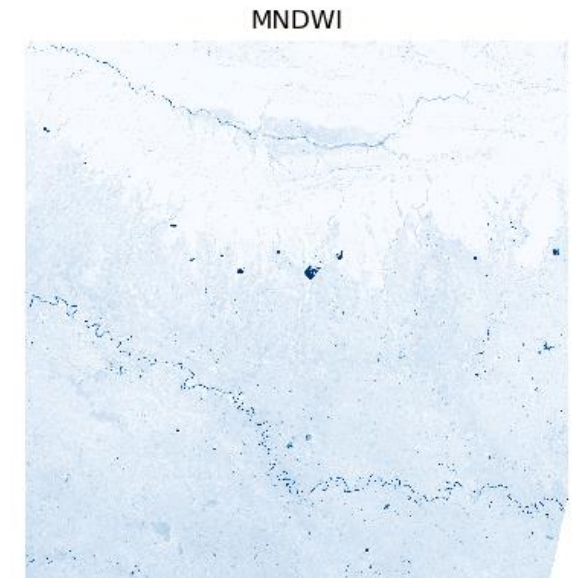
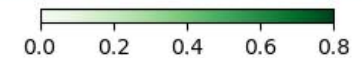
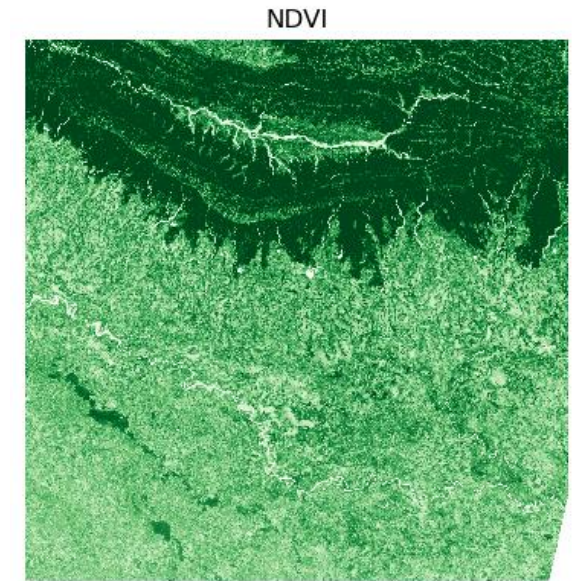
(a)

(b)

Wang et al. 2023

Spectral indices

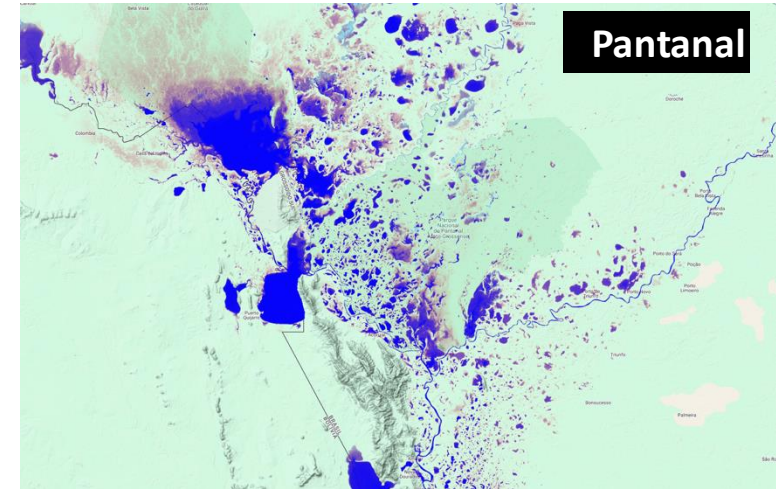
- Spectral indices serve as proxies for some environmental characteristic of interest
 - Vegetation productivity
 - Moisture content
 - Burned areas
 - Human development
- Involve the use of bands from the multispectral sensors (typically optical)



Common variables used in movement ecology

- Varies by environment
 - Terrestrial
 - LULC, elevation, slope, bioclimatic variables, vegetation indices, distance to roads and water
 - Marine
 - Depth, SST (and SD), SSH (and SD, EKE, Chl a, SSS, currents

JRC Global Surface Water



CMEMS; GLORYS

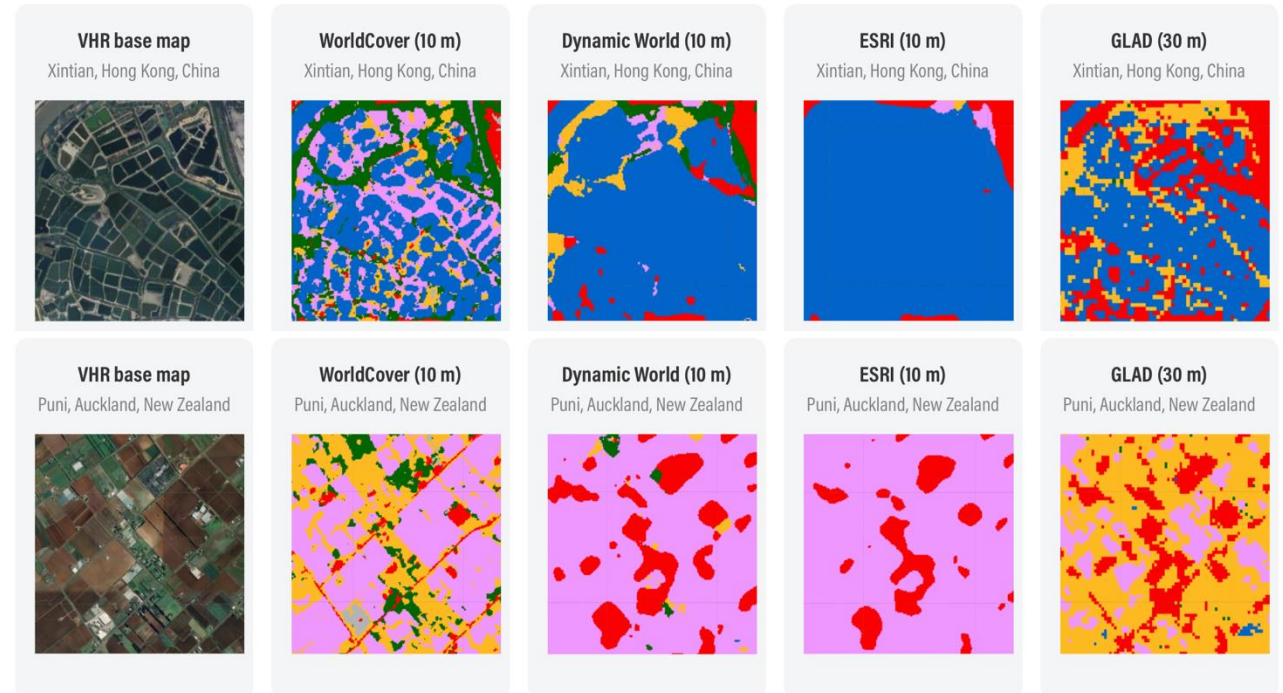
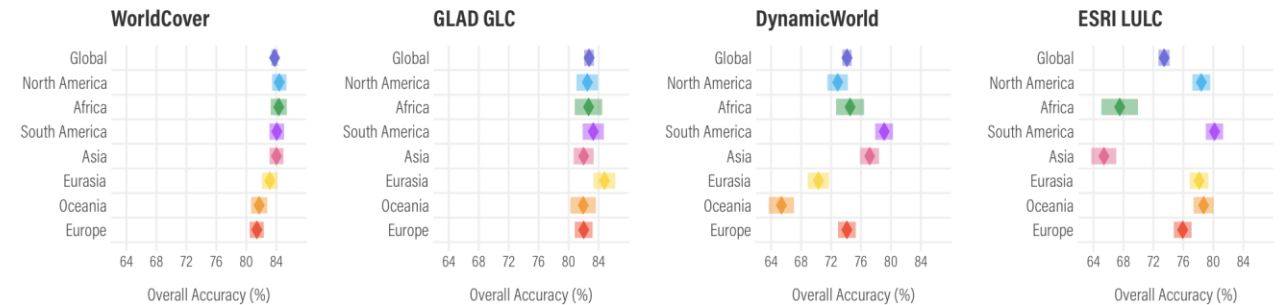
Comparing global LULC products

- Based on paper by Xu et al. (2024)
 - And associated blog post by Carter et al. (2024)

- Main takeaways are:

1. classification accuracy varies by product and continent
2. 30 m products can still perform as well or sometimes better than 10 m products
3. ESA WorldCover and GLAD GLC performed best overall

Overall accuracy of the global land cover maps by continent



Legend Water Trees Mixed vegetation Cropland Built areas Bare ground Snow & Ice

Comparing global LULC products

Global land cover map fitness for purpose

● Data set is best for this purpose ● Data set can serve this purpose but another data set is better ● Data set cannot be used for this purpose

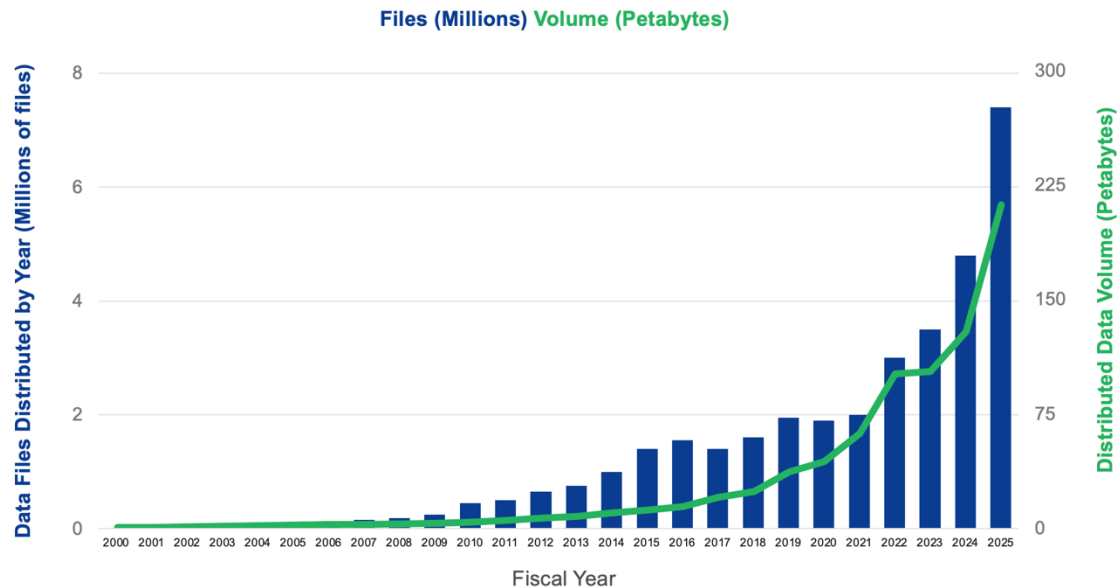
Purpose	WorldCover	Dynamic World	ESRI LULC	GLAD GLC
Use data spanning several years	●	●	●	●
Derive sub-annual land cover changes, and user specified maps	●	●	●	●
Integrate with Global Forest Watch tree cover and change products	●	●	●	●
View some 10-meter spatial detail	●	●	●	●
Easily derive annual information	●	●	●	●
Fine tune legend for user specific maps using thematically detailed classes	●	●	●	●
Gain insight into mixed classes due to probability data, or per pixel land cover fractions	●	●	●	●
Identify grassland	●	●	●	●
Identify urban green spaces	●	●	●	●
Identify crop areas	●	●	●	●
Identify wetland areas	●	●	●	●
Identify mangroves	●	●	●	●

The rise of Big Data

- With petabytes of data comes need for ways to process all of this

NASA-only data

Total Data Volume and Data Files Distributed by Year



 Google Earth Engine

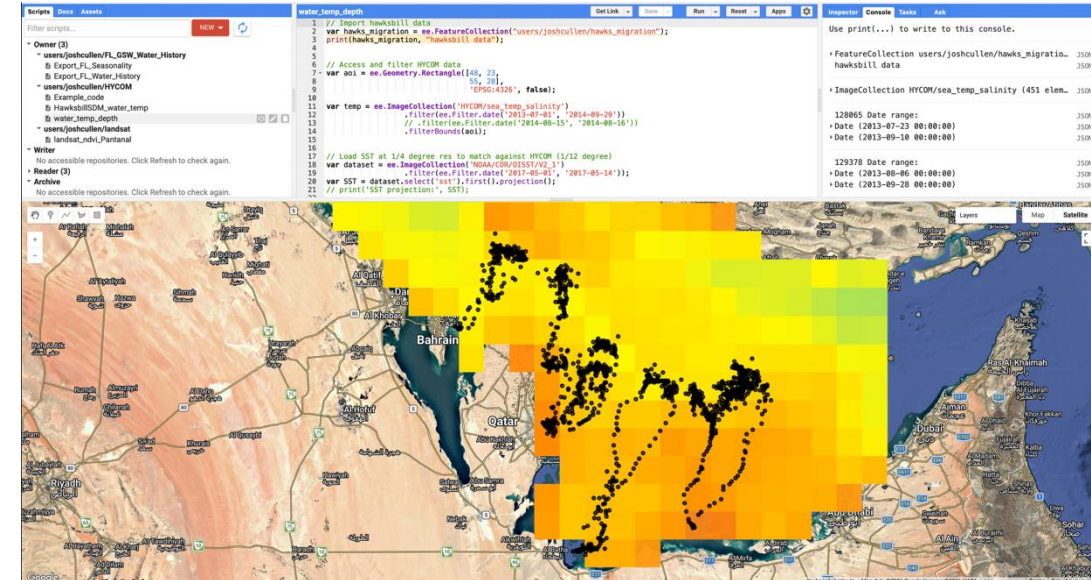


Cloud-optimized Geotiff (COG)
Zarr

Google Earth Engine (GEE)



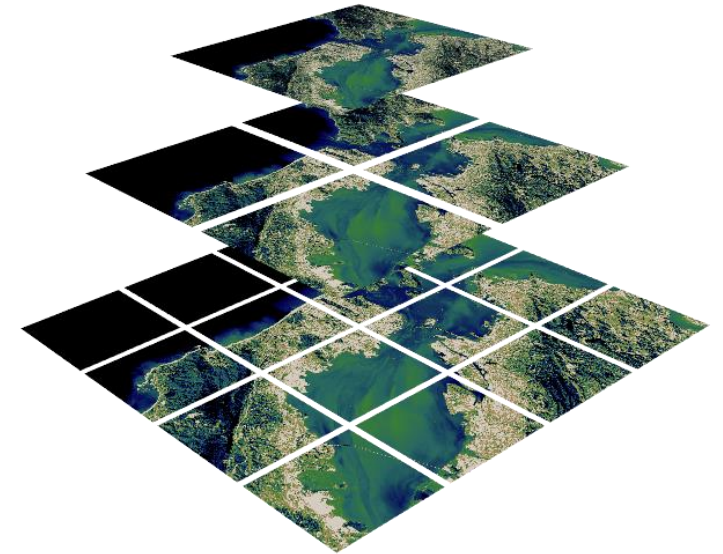
- Likely most familiar and widely used platform for cloud-processing of remote sensing data
 - All geoprocessing done on Google servers instead of your computer
 - No more limitations in memory/storage of your PC!
- Hosts wide-array of datasets for access and processing
- Primary ways to access and visualize data are through the GEE Code Editor using JavaScript, or through the Python API
- {rgee} R package available, but requires working through Python



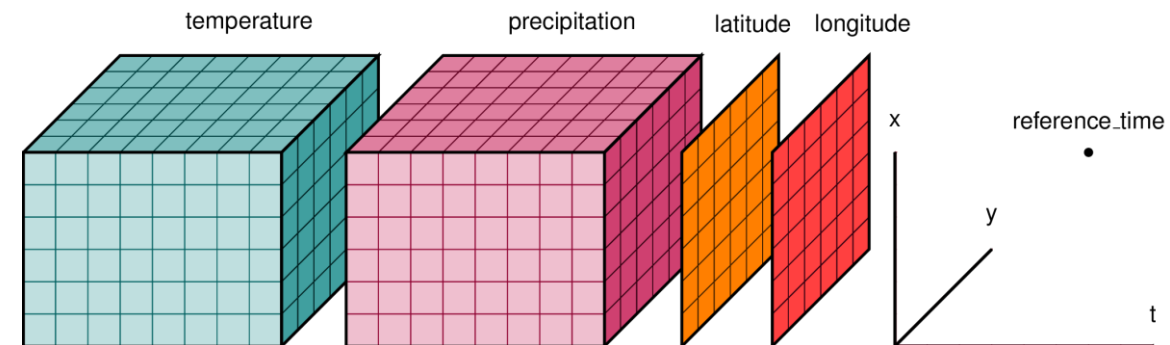
Data from Marshall et al., 2020

Cloud-native raster data formats

- Allow lazy loading (via reading of metadata only)
- Select range of data (over space and/or time)
- Allows for partial and parallel reads of data chunks
- More memory efficient, especially when data is not local



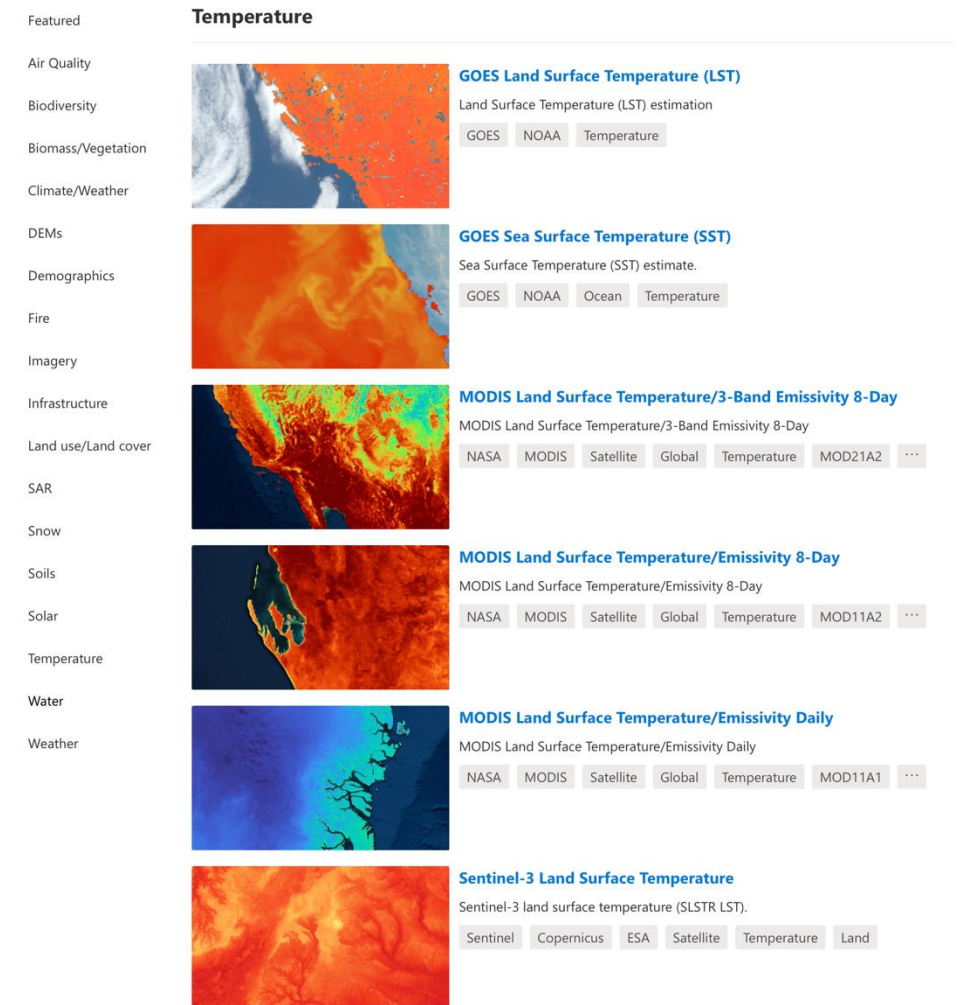
<https://www.kitware.com/deciphering-cloud-optimized-geotiffs/>



<https://xarray.dev>

Major sources of remote sensing data

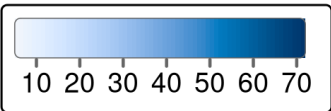
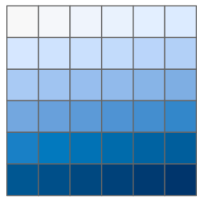
- Terrestrial
 - STAC Index
 - GEE
 - Microsoft Planetary Computer
 - Digital Earth Africa
 - NASA Earthdata
 - Source Cooperative
 - Copernicus Land Monitoring Service
- Marine
 - NOAA ERDDAP
 - Copernicus Marine Environmental Monitoring Service
 - NASA Earthdata



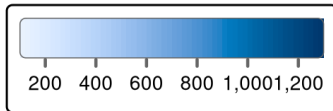
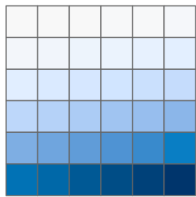
Accessing and processing data in R

- We can use `{terra}` or `{stars}` to read most cloud-based or local raster data
 - I'll focus on `{terra}`

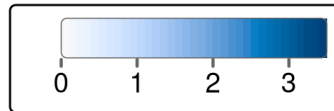
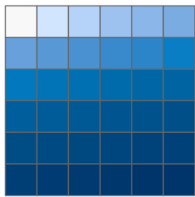
`elev + elev`



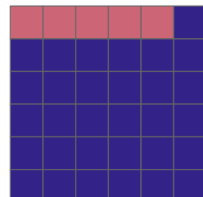
`elev^2`



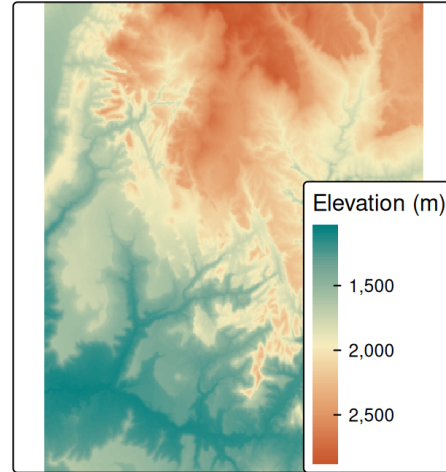
`log(elev)`



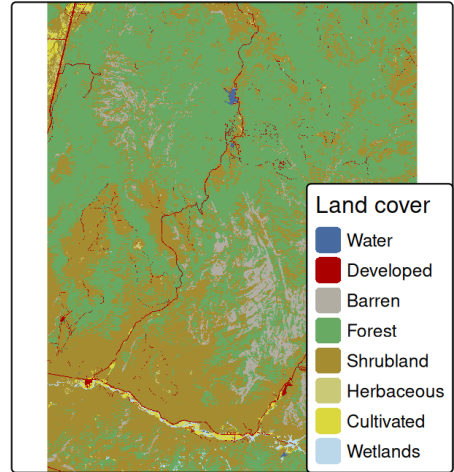
`elev > 5`



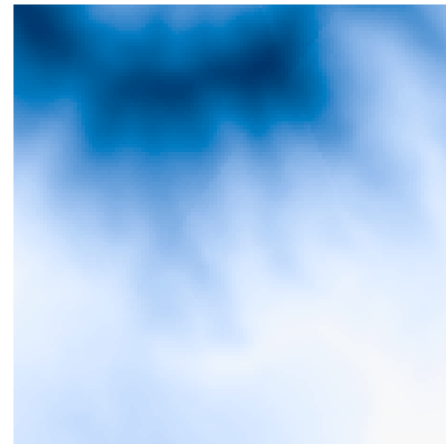
A. Continuous data



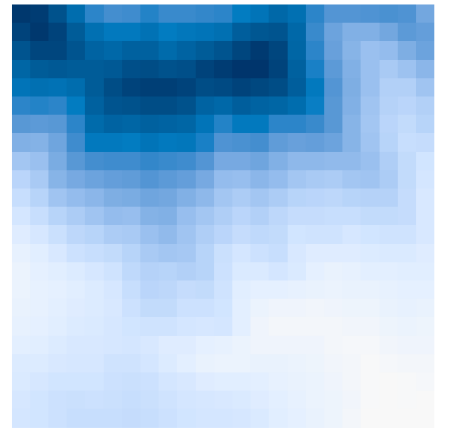
B. Categorical data



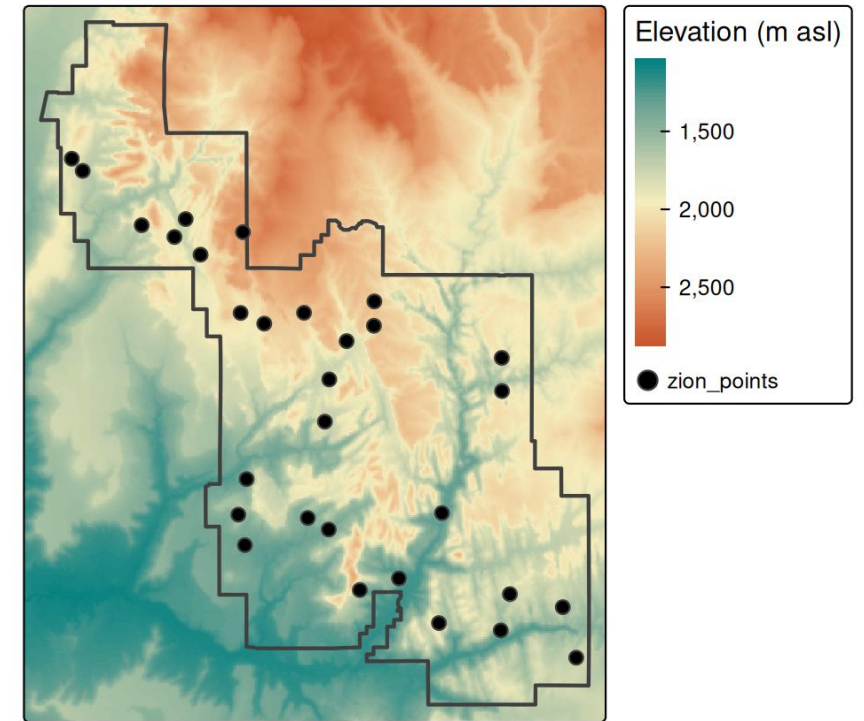
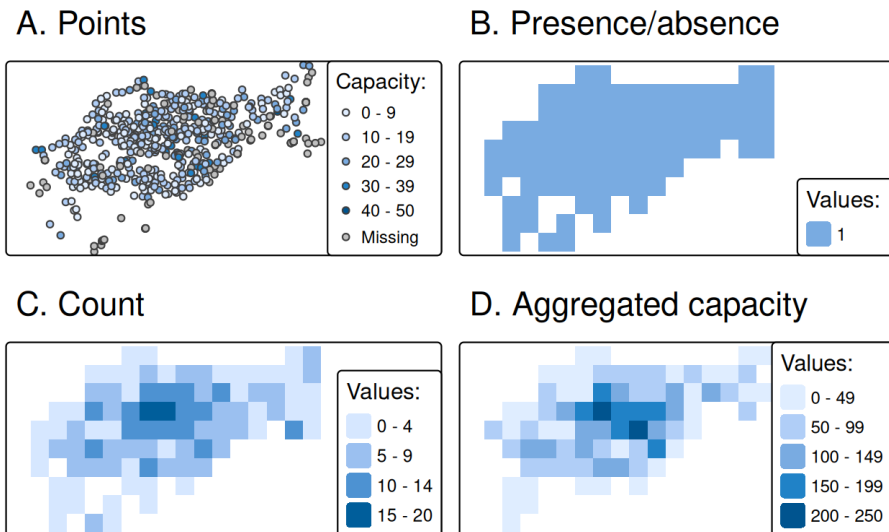
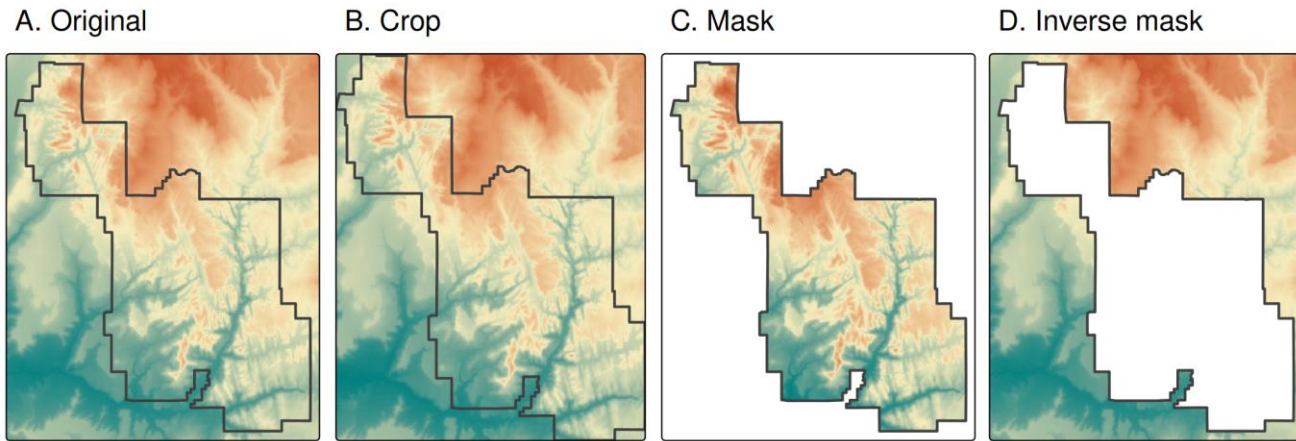
A. Original



B. Aggregated

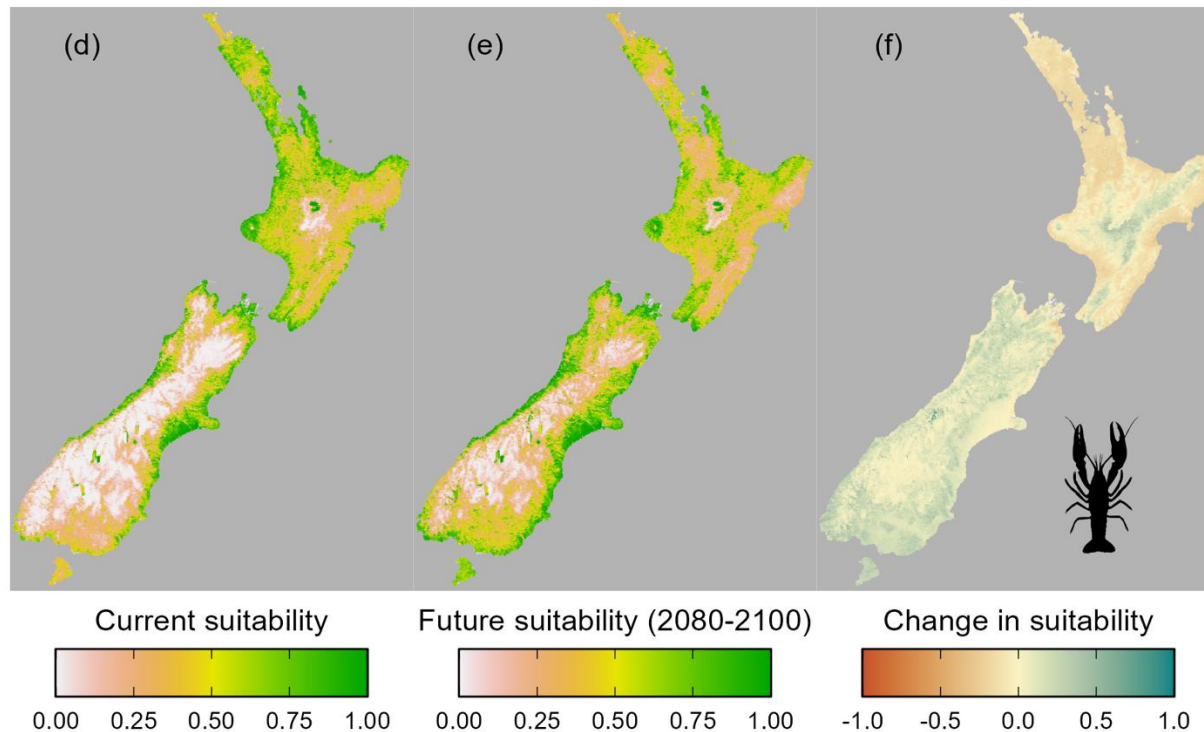


Accessing and processing data in R

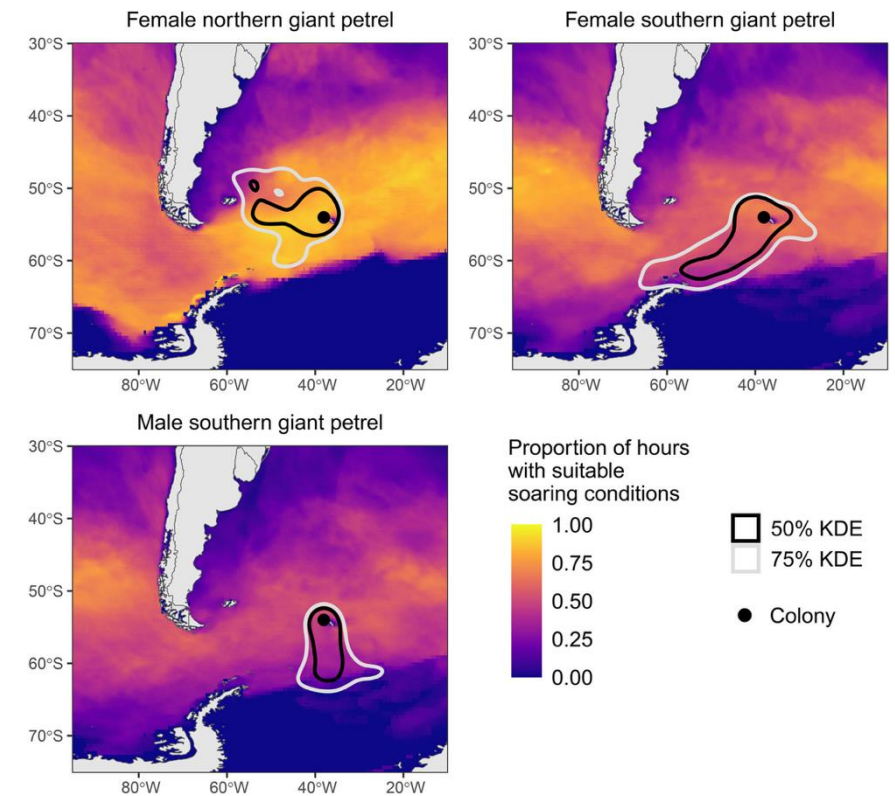


Visualizing data in R

- There are several options for static maps, but I like to use `{ggplot2}` and `{tidyterra}`



Lee et al. 2025



Hallet et al. 2026

Time for live-coding



Download and process remote sensing data, manage large datasets, extract layers at telemetry locations